ECE 4331 – 102

Project Marketing Engineering Internship Proposal

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1. Abstract

This proposal describes the anticipated technical work to be completed during a Product Marketing Engineering internship within the Low Power Connectivity Solutions organization at Texas Instruments. The internship will focus on developing knowledge and experience in wireless embedded systems, including Bluetooth Low Energy (BLE), BLE Mesh networking, Zephyr Real-Time Operating System (RTOS), over-the-air (OTA) update concepts, and low-power connectivity solutions. Initial activities have included familiarization with the CC2340R53 wireless microcontroller, BLE demonstrations, and investigation of TI's software development tools and mobile applications.

The primary objective of this internship is to apply technical knowledge to create demonstrations and educational resources that support developer understanding and adoption of Texas Instruments wireless technologies. The work will require collaboration with multidisciplinary engineering teams and may expand to include additional technical assignments as organizational priorities evolve throughout the internship. Through a combination of technical investigation, hands-on development, validation, and technical communication, this experience provides a challenging opportunity to strengthen embedded systems knowledge while addressing real-world engineering needs within the wireless connectivity industry.

2. Background

The rapid growth of Internet of Things (IoT) devices has increased the demand for low-power wireless communication technologies capable of providing reliable connectivity while minimizing energy consumption. Technologies such as Bluetooth Low Energy (BLE) are widely used in wearable electronics, consumer products, industrial monitoring systems, medical devices, and automotive applications due to their ability to support efficient data transfer with low power requirements. As the number of connected devices continues to increase, engineers must develop solutions that are both technically effective and accessible to developers implementing these technologies.

In addition to advances in wireless hardware, embedded software ecosystems have evolved to simplify application development and accelerate product deployment. The Zephyr Real-Time Operating System (RTOS) has become an increasingly popular open-source platform for embedded applications because of its broad hardware support and extensive software features. Developers frequently rely on technical demonstrations, tutorials, documentation, and example applications to evaluate new technologies and learn implementation workflows. As a result, developer enablement has become a key component of promoting the successful adoption of embedded systems solutions.

Texas Instruments' Low Power Connectivity Solutions organization develops wireless technologies that support a wide range of applications requiring dependable, energy-efficient communication. Within this organization, Product Marketing Engineers work closely with applications engineers, systems engineers, and other technical teams to communicate complex engineering concepts to customers and developers through demonstrations, educational materials, and technical presentations. This internship provides the opportunity to gain experience with

wireless embedded systems while contributing to developer-facing resources that support the understanding and adoption of Texas Instruments connectivity solutions.

3. Objectives

As a Product Marketing Engineering Intern, I will have the opportunity to explore emerging wireless technologies while bridging the gap between technical implementation and developer education. This experience will allow me to build confidence in wireless embedded systems concepts, collaborate with multidisciplinary engineering teams, and apply engineering knowledge in ways that help others understand and adopt Texas Instruments connectivity solutions.

3.1 Develop proficiency in wireless communication technologies

A major objective of this internship is to gain a deeper understanding of wireless communication concepts relevant to embedded systems applications. This includes investigating Bluetooth Classic, Bluetooth Low Energy (BLE), BLE Mesh networking, low-power connectivity principles, and the role these technologies play in modern Internet of Things (IoT) systems. The knowledge gained through this investigation will provide a stronger foundation in wireless communication systems and their practical implementation.

3.2 Gain experience with embedded development tools and workflows

This internship will provide firsthand exposure to Texas Instruments hardware and software ecosystems used in low-power wireless applications. Responsibilities include becoming familiar with the CC2340R53 wireless microcontroller, evaluating example applications, utilizing TI development tools and mobile applications, and investigating development workflows involving the Zephyr Real-Time Operating System (RTOS) and over-the-air (OTA) update concepts. Initial activities have already included exploring BLE examples and becoming familiar with the functionality of TI wireless platforms.

3.3 Develop technical demonstrations and educational resources

Another objective is to apply acquired technical knowledge toward the development of

demonstrations and educational materials intended for developer audiences. These resources may include technical presentations, instructional videos, example walkthroughs, and supporting collateral designed to improve understanding of TI connectivity solutions and simplify the adoption of wireless technologies.

3.4 Collaborate with multidisciplinary engineering teams

The internship provides opportunities to interact with engineers representing multiple technical disciplines and product areas. Collaboration with applications engineers, systems engineers, product marketing engineers, and teams supporting 2.4 GHz, Sub-1 GHz, and automotive connectivity solutions will provide insight into how engineering decisions are communicated across organizations and translated into customer-facing resources.

3.5 Contribute to future technical assignments throughout the internship

Because the internship extends beyond the initial project activities, an additional objective is to support future assignments involving low-power connectivity technologies as organizational priorities evolve. These responsibilities may include investigating emerging technologies, developing additional demonstrations, supporting technical marketing initiatives, or contributing to new projects introduced during the remaining duration of the internship. This objective ensures continued technical growth and adaptation throughout the internship experience.

4. Methodology

The approach taken throughout this internship emphasizes continuous learning through investigation, experimentation, validation, and collaboration while adapting to evolving project responsibilities. By combining technical exploration with practical application, this methodology supports the development of wireless embedded systems knowledge while contributing to resources that help developers understand and adopt Texas Instruments connectivity solutions.

4.1 Technical Investigation and Knowledge Development

The first phase of the internship involves developing an understanding of wireless communication technologies relevant to Texas Instruments' Low Power Connectivity Solutions portfolio. This includes investigating Bluetooth Classic, Bluetooth Low Energy (BLE), BLE Mesh networking, Zephyr Real-Time Operating System (RTOS), over-the-air (OTA) update concepts, and low-power connectivity principles through technical documentation, internal resources, presentations, and discussions with engineering teams.

4.2 Embedded Development and Demonstration Activities

Following the investigation phase, direct activities will be used to reinforce technical concepts and familiarize the intern with TI hardware and software ecosystems. These activities include evaluating example applications using the CC2340R53 wireless microcontroller, exploring Texas Instruments development tools and mobile applications, modifying existing BLE examples, and validating system functionality through demonstration exercises. Preliminary activities have included BLE advertising demonstrations and embedded software familiarization to establish a foundation for future assignments.

4.3 Validation and Technical Review

Demonstrations and example implementations will be evaluated to ensure proper

functionality and technical accuracy. Validation activities may include confirming expected software behavior, reproducing implementation workflows, reviewing technical content with subject matter experts, and incorporating feedback from engineering teams. This process will help ensure that the resulting demonstrations and educational materials accurately represent the capabilities of TI connectivity solutions.

4.4 Development of Educational Resources

Technical knowledge acquired throughout the internship will be translated into developer-facing resources intended to improve understanding and adoption of TI wireless technologies. Deliverables may include instructional videos, technical presentations, example walkthroughs, and supporting collateral designed to communicate complex engineering concepts in an accessible manner while maintaining technical integrity.

4.5 Expansion to Future Technical Assignments

As the internship progresses and initial responsibilities are completed, additional projects may be assigned based on organizational priorities and emerging technical needs. Future activities may involve investigating new low-power connectivity technologies, supporting additional demonstrations, contributing to technical marketing initiatives, or participating in other engineering-related assignments introduced during the remainder of the internship. This approach allows for continuous technical growth and adaptation throughout the internship experience.

References

1. Bluetooth Special Interest Group, "Bluetooth Technology Overview," Bluetooth SIG, https://www.bluetooth.com/learn-about-bluetooth/tech-overview/ (accessed: June 10, 2026).
2. Texas Instruments, "Bluetooth Low Energy Fundamentals," Texas Instruments, https://dev.ti.com/tirex/explore/node?node=ADoEZBrWKWqhpyfpKc8WDw__krol.2c__LATEST (accessed: June 10, 2026).
3. Texas Instruments, "TI Bluetooth Mesh Fundamentals," Texas Instruments, https://dev.ti.com/tirex/explore/node?node=A__ASi6dg7aXrVhEGhEdi8ISg__com.ti.SIMPLELINK_ACADEMY_CC13XX_CC26XX_SDK__AfkT0vQ__LATEST (accessed: June 10, 2026).
4. Texas Instruments, "Zephyr Project Environment Setup - SimpleLink Academy for CC23xx," Texas Instruments, https://dev.ti.com/tirex/explore/node?node=A__Abn1NAQObvrVu7R5iV50Lw__SIMPLELINK-ACADEMY-CC23XX__gsUPh5j__LATEST (accessed: June 10, 2026).
5. Texas Instruments, "CC2340R SimpleLink™ Family of 2.4GHz Wireless MCUs Datasheet," Texas Instruments, https://www.ti.com/lit/ds/symlink/cc2340r5.pdf (accessed: June 10, 2026).
6. Texas Instruments, "LP-EM-CC2340R53 LaunchPad Development Kit for CC2340R53," Texas Instruments, https://www.ti.com/lit/ug/swru637/swru637.pdf (accessed: June 10, 2026).